DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF SCIENCE AND COMPUTER STUDIES
INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS

COURSE CODE	**:** S-ITPC227
COURSE TITLE	**:** Application Development and Emerging Technologies
COURSE TYPE	**:** Laboratory
COURSE CREDIT	**:** One Unit
LEARNING MODALITY	**:** Fully Onsite
PRE-REQUISITES	**:** None
CO-REQUISITES	**:** S-ITPC227LA Application Development and Emerging Technologies Lab
PROFESSOR	**:** MR. JUANITO C. DOCTOR, JR.
CONSULTATION HOURS	**:** Monday 0700-1000 / COS100C Wednesday 1100-1200 / COS100C Wednesday 0800-1100 / Online Friday 1100-1200 / Online
CONTACT DETAILS	**:** jcdoctor@dlsud.edu.ph (02) 7795180 □ (046) 4811900 local 3134

VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

This course focuses on the development of applications using web, mobile and emerging technologies with emphasis on requirements management, interface, design, usability, testing and deployment, including ethical and legal considerations.

COURSE LEARNING OUTCOMES:
By the end of this course, students can:

CLO1. interpret the difference of front-end and back-development in web application;
CLO2. understand Web Development concept by dealing with client-based case study;
CLO3. implement Object Oriented way of programming;
CLO4. Understand how to manage website using Model, View and Control;
CLO5. develop and deploy complete Web Application in live server for pilot testing; and
CLO6. demonstrate 21st century skills in all learning activities.

LEARNING PLAN:

Learning Outcomes for Midterm Period

<table>
<tr><th>Course Learning Outcomes</th><th>Topic Learning Outcomes</th></tr>
<tr><td>CLO1.Interpret the difference of front-end and back-development in web application.</td><td>TLO1. Learn and differentiate different web development software and its features.

TLO2. Obtain tools in developing Web Application Project. Students can implement objects in their codes.</td></tr>
<tr><td>CLO2. Understand Web Development concept by dealing with client-based case study.</td><td rowspan="2">TLO3. Design web application both front-end and back-end. Students can know and use different OOP programming techniques.</td></tr>
<tr><td>CLO3. Implement Object Oriented way of programming.</td></tr>
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Module No and Learning Outcomes	Week	Teaching-Learning Activities/Assessment Strategy	Technology Enabler	Allotted Hours
Module 0	**Week 1** Jan 20 – Jan 25	• Gospel Reading, Reflection and Prayer • Demonstration of Course Syllabus **Module 0: Course Introduction** • Introduction to web application	Schoolbook, PowerPoint	3
Module 1 CLO1 TLO1 TLO2	**Week 2** Jan 27 – Feb 1	• Gospel Reading, Reflection and Prayer • Demonstration of: **Module 1: Understanding client-side and server side controls** • Enabling Assessment: Creating Simple Web Application	Schoolbook, PowerPoint, Visual Studio .NET	3
Module 1 CLO1 TLO1 TLO2	**Week 3** Feb 3 – Feb 8	• Gospel Reading, Reflection and Prayer Enabling Assessment: Hands-on exercise on Implementing Web application using C# as a platform	Visual Studio .NET	3
Module 2 CLO2 TLO2 TLO3 TLO4	**Week 4** Feb 10 – Feb 15	• Gospel Reading, Reflection and Prayer • Demonstration of: **Module 2: Implementing Object Oriented programming in Web application**	Schoolbook, PowerPoint, Visual Studio .NET	3

LONG EXAM	**Week 5** Feb 17 – Feb 22	• Gospel Reading, Reflection and Prayer Demonstration Enabling Assessment: Hands- on Exercise on Implementing OOP Technique in web application	Visual Studio .NET	3
Module 3 CLO2 TLO3 TLO4 TLO5	**Week 6** Feb 24 – Mar 1	• Gospel Reading, Reflection and Prayer • Demonstration of **Module 3: Introduction to Master Page** • Enabling Assessment: Hands- on Exercise on Integrating Master Page in Web application	Schoolbook, PowerPoint, Visual Studio .NET	3
Module 3 CLO3 TLO5 TLO6 TLO7	**Week 7** Mar 3 – Mar 8	• Gospel Reading, Reflection and Prayer • Enabling Assessment: (A 40 points hands-on using different client-side control with application of Master page)	Schoolbook, PowerPoint, Visual Studio .NET	3
Module 1-3	**Week 8** Mar 10 – Mar 21	• Midterm Summative Assessment on Modules 1-3		3
TOTAL				24

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO4. Manage website using Model, View and Control.	TLO8. Discuss the principle of Model, View and Control in mobile application TLO9. Discuss advance android controls. TLO10. Introduction to Report module
CLO5. Develop and deploy complete Web Application in live server for pilot testing.	TLO11. Implementing Report data visualization commands. TLO12. Deploying website to a live server.
CLO6. Demonstrate 21st century skills in all learning activities.	*Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-world Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcomes	**Date**	**Teaching-Learning Activities /Assessment Strategy**	**Technology Enabler**	**Allotted Hours**
Module 4 CLO4 TLO8 TLO9 TLO10	**Week 10** Mar 24 – Mar 29	• Gospel Reading, Reflection and Prayer • Demonstration of Module 4 • Enabling Assessment 1: Laboratory Exercise on Program Simulation of Module 4 • Sustainability Week	Schoolbook, PowerPoint, Visual Studio .NET	3

Module 4 CLO4 TLO8 TLO9 TLO10	**Week 11** Mar 31 – Apr 5	• Gospel Reading, Reflection and Prayer • Demonstration of **Module 4: Introduction to ADO.NET (SQL Server) Database application** • Enabling Assessment: Case Study on creating a web application with database application	Schoolbook, PowerPoint, Visual Studio .NET	3
Module 4 CLO4 TLO8 TLO9 TLO10	**Week 12** Apr 7- Apr 12	• Gospel Reading, Reflection and Prayer Enabling Assessment: Hands-on on Implementing SQL Commands in Web Application	Schoolbook, PowerPoint, Visual Studio .NET	3
	Week 13 April 14-19	Independent Learning Week (April 14-15) Holy Week (16-19)		3
	Week 14 Apr 21-Apr 26	Research Week		3

Module 5 CLO5 TLO11 TLO12	**Week 15** Apr 28-May 3	• Gospel Reading, Reflection and Prayer • Demonstration of Module 5 • Enabling Assessment: Case study on Implementing Stored procedure	Schoolbook	3
Module 6 CLO5 CLO6 TLO11 TLO12	**Week 16** May 5 – May 10	• Gospel Reading, Reflection and Prayer • Demonstration of **Module 6: Using Reporting tool in Web appliction** • Enabling Assessment: Case Study on Implementing Deploying web application on a live server	Schoolbook, PowerPoint, Visual Studio .NET	3
	Week 17 May 12 – May 17	• Final Term Summative Assessment on Module 4-4		3
TOTAL				**24**

GRADING SYSTEM

Midterm	%	**Final**	%
Enabling Assessment	40%	**Enabling Assessment**	40%
Summative Assessment	60%	**Summative Assessment**	60%
	100%		100%

FINAL GRADE = $\frac{\text{Midterm Grade + Final Grade}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
    a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
    b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
        i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
        ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
        iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
    c. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
    a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for student's grade.
    b. The onsite meeting will only be held at the schedule indicated in the registration form.
    c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.

d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
    b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
    c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones cannot also be used as calculator during examination.
    d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES

Onsite References

Provider	**Reference Material**
AEA Library QA 76.73.C154 .N131	Nagel C. 2019. Professional C# 6 and .Net Core 1.0: Indianapolis, Indiana Wrox, a Wiley brand
AEA Library QA 76.73.C154 .Se54	Sempf B. Sphar C. Davis,SR. 2019. C# 5.0 all-in-one for dummies

Online References

Call Number/ E-provider	Reference Material

Web resources	C# Tutorials https://www.tutorialspoint.com/csharp/index.htm
Web resources	What is ASP.NET? https://youtu.be/3AYoipyqOkQ?list=PL6n9fhu94yhXQS_p1i-HLIftB9Y7Vnxlo
Web resources	ASP .NET Tutorials https://learn.microsoft.com/en-us/aspnet/tutorials

Prepared: **Information Technology Department**
2nd Semester 2024-2025

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